

STANDARD OPERATING PROCEDURE:

TEMPERATURE DATA PROCESSING FOR THE TERRITORIAL CORAL REEF MONITORING PROGRAM

INTRODUCTION

The United States Virgin Islands (USVI) Territorial Coral Reef Monitoring Program (TCRMP) is administered by the Department of Planning and Natural Resources and conducted by the Center for Marine and Environmental Studies and the University of the Virgin Islands. The purpose of the program is to provide critical information on the status and threats to all Virgin Islands coral reef ecosystems in order to increase management effectiveness and improve basic and applied coral reef research.

Temperature is a fundamental measure of the energy. Its values are critical in the ecology of coral reef ecosystems, affecting coral thermal stress, organism reproduction, and disease incidence. Background sea surface temperatures and incidence of coral thermal stress have been increasing in the world's oceans and the USVI as a result of human-induced climate change and global warming. The recent rate of warming in the USVI has been $\sim 0.2^{\circ}\text{C}$ per decade (Figure 1) which has contributed increasing frequency of annual high temperature deviations above the local coral thermal bleaching threshold of 29.5°C , of accumulation of heat stress (degree heating weeks), and of mass coral bleaching and related coral disease and mortality.

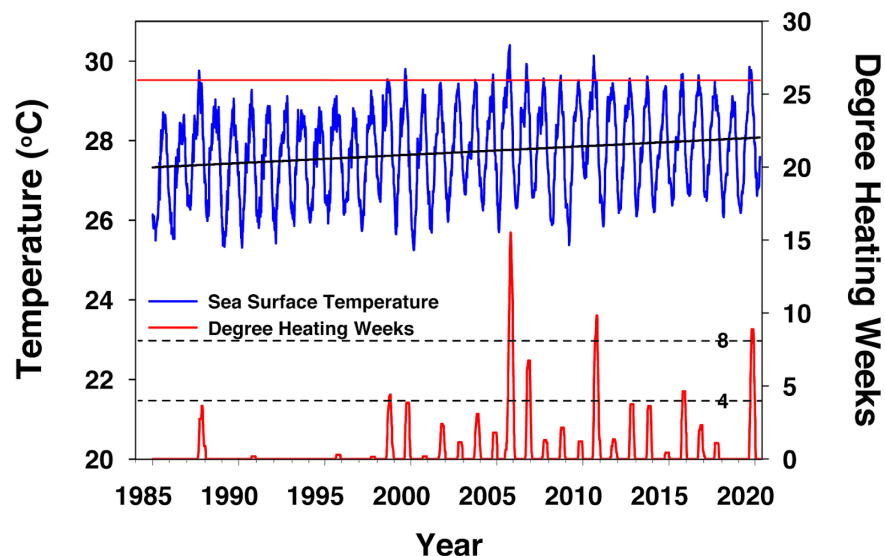


Figure 1. Sea surface temperatures and degree heating weeks of the US Virgin Islands from 1984-2020. Temperatures have been increasing at about 0.2°C per decade across this record, leading to increasing frequency and severity of coral bleaching events. Degree heating weeks take into account the duration and severity of heating above the bleaching threshold temperature (red line at 29.5°C). Degree heating weeks above 4 are associated with the onset of coral bleaching and above 8 are associated with mass bleaching and the start of coral mortality. Data from NOAA Coral Reef Watch 5 km product (<https://coralreefwatch.noaa.gov/satellite/ge/index.php>; accessed April 20, 2020).

In addition, temperature is highly variable with geographic location and with depth. Therefore, monitoring of near reef temperatures across coral reef monitoring sites is critical to understanding the status and threats to coral reefs of the USVI and informing effective management.

For this reason, the TCRMP maintains temperature monitoring stations at more than 40 locations throughout the USVI from depth of 1 – 100 m depth. It is imperative that this information be *reliably collected in the field, updated annually, quality controlled, well described, reliably stored for future retrieval, and made available to any person or organization for informational and research purposes*. The purpose of this SOP is for the processing and maintenance of TCRMP temperature data and metadata in a consistent and reliable set of formats.

PROCESS DESCRIPTION

Temperature data are typically taken with HOBO Water Temp Pro v2 sensors (Onset Computer Corp., Bourne, MA) tied with plastic cable ties on steel rods fixed to the seafloor (Figure 2). Probes are emplaced by divers. At continuous monitoring sites the loggers are swapped for new or reset calibration controlled loggers. Loggers are checked prior to deployment for accuracy in a freshwater ice bath and not used if temperatures deviate more than 0.2°C from the mean of multiple loggers or from the actual temperature of the ice bath (see Appendix A for more details on calibration check method).



Figure 2. A Water Temp Pro V2 probe pair affixed by a diver at the Salt River 67 m depth TCRMP site. (Credit: V. Brandtneris, Date: November 4, 2018)

Loggers retrieved from the field are transported back to the lab, cleaned, and the data is offloaded to computers using an optical base station. Loggers start date and time can be delayed to coincide with when they are expected to be deployed; however, it is normal for unprocessed files to include temperature readings at the beginning and end of the record that were recorded in air and before the loggers were in their intended location of deployment. These need to be trimmed off the final QA/QC data file.

Also, in some sites there may be more than one logger put in place to provide a duplicate record for high-value time series or to allow a longer deployment by offsetting the recording time intervals (memory storage on the loggers limits the duration of the deployment). For duplicate loggers, if the sensors are judged to have been recording reliable and accurate data, then the multiple files need to be merged into a single file.

Finally, each deployment must be assigned with metadata that describes the who, what, where, when of the deployment. Some details are input by the person conducting the data processing and some technical details are available for downloading from the logger.

The result of this process are four files: The data file directly from the logger (typically .hobo format), a text details file of the deployment metadata from the logger, a text metadata file containing all information from the deployment, and a QA/QC temperature file that contains the temperature data [comma separate values (.csv) format; Figure 3].

	A	B	C	D
1	Number	Date Time UTC 4	Temperature	
2	1	9/4/07 16:22	29.315	
3	2	9/4/07 16:37	29.315	
4	3	9/4/07 16:52	29.29	
5	4	9/4/07 17:07	29.265	
6	5	9/4/07 17:22	29.24	
7	6	9/4/07 17:37	29.215	
8	7	9/4/07 17:52	29.215	
9	8	9/4/07 18:07	29.215	
10	9	9/4/07 18:22	29.215	
11	10	9/4/07 18:37	29.24	
12	11	9/4/07 18:52	29.24	
13	12	9/4/07 19:07	29.215	
14	13	9/4/07 19:22	29.215	
15	14	9/4/07 19:37	29.215	
16	15	9/4/07 19:52	29.24	
17	16	9/4/07 20:07	29.24	
18	17	9/4/07 20:22	29.215	
19	18	9/4/07 20:37	29.215	

Figure 3. Example of a Quality Assured-Quality Controlled temperature file with standardized data headers (row 1, columns a:c). UTC 4, stands for Coordinated Universal Time – 4 hours.

STANDARD OPERATING PROCEDURE

KEY FILES

TEMPERATURE_UVI_DEPLOYMENT_LOG.XLSX

This file records the status of loggers and deployments. This is a key file used during the creation of the file metadata, naming of files, and in trimming the data to the actual deployment.

Data includes (by column):

Status =	_____	Is the logger running, removed, or lost
Location =	_____	Location as the full name of the site
Logger =	_____	The specific logger indicated by its serial number
Tag ID =		If given an external tag in the field (historical logger deployments)
Date On =	_____	The date the logger started recording data
Frequency =	_____	The logging frequency set by the user. Units in minutes
Duration =		The expected time until logger full set by the sampling frequency. Units in days
Date In =	_____	Date that the logger was deployed in the field at the final sampling site
Time In =	_____	Time that the logger went in the field at the final sampling site
Date Full =	_____	The date the logger memory will be full based on Duration and Date On
Date Offloaded =		The date the logger was stopped from recording and data offloaded from the logger
Offloaded File Name =	_____	The file name given to the offloaded file and put in the folder for processing
Date Out =	_____	The date the logger was retrieved from the site in the field
Time Out =	_____	The time the logger was retrieved from the site in the field
Notes =	_____	Anything of importance pertaining to the deployment or file

RAW DATA FILE DESCRIPTION

All field downloaded temperature data files (.hobo) and exported spreadsheet files (.csv) are deposited into a single folder. Each file is named to identify the data type, project, site, the deployment year-month, and the number of the logger (if more than one at a site).

This file name serves as an example: **BT_TCBKPT_1811_a.hobo**

Where,

BT = Benthic Temperature, or the type of data set

TC = TCRMP, or the specific project

BKPT = The four letter location code. In this case Black Point. Site codes are fixed and can be found on the Site Code Metadata (Appendix B)

1811 = The deployment year and month coded as YYMM, in this case November 2018

a = The logger number. This site has two loggers coded as “a” and “b”. This is logger “a”

STEP 1: DOWNLOADING THE DATA FILES (AND MERGING FILES)

In the folder with the raw offloaded temperature files locate (1) the .csv file and (2) the .hobo file corresponding to the same date at a certain location.

Go to “TCRMP-Temperature_Benthic_SiteList&Progress_EXAMPLE” excel sheet in the Temperature Data Processing Folder. Create your own sheet with your initials at the end and copy/paste original hobo file name into the column labeled “Source File”.

If there are two loggers for the same site and deployment dates, record each of these in separate rows. In addition, these files should be merged and checked before columns are renamed and trimmed using this procedure:

Merging files: For some sites, there are 2 hobo files corresponding to the same site and date. Determine if the loggers are duplicates or offsets.

Duplicates recorded a temperature reading at the same time and were deployed at high value information sites to maintain temperature records in the event of a logger failure or loss. These temperatures need to be averaged at each date/time point if they meet certain criteria outlined below.

Offsets recorded temperatures in an alternating way and were used to extend logger deployments and provide some backup data in the event of a logger failure. For example loggers might record every 30 minutes, but logger “a” records on the hour and half hour, while logger “b” records on the hour’s fifteen and forty five minute mark. Merged, these files would produce a continuous file with readings every :15 minutes, but with a deployment duration that is twice as long as a single logger with a :15 recording frequency. These loggers are used in hard to access areas. These data need to be merged into a single file with readings sequential.

Duplicate example. For example, at the site Black Point there is a .csv and .hobo file each named “BT_TCBKPT_1509_a” as well as a csv and hobo file each named “BT_TCBKPT_1509_b”.

Process them as one set. The “a” vs. “b” correspond to the different serial numbers of the loggers (two different devices were put into the water at the same time). In this case, copy and paste the temperatures from the csv files corresponding to both loggers and put them in the same excel sheet (make sure that the dates/times of the recorded temperatures line up.) Then subtract them from one another to find the difference (absolute value). If the difference is 0.2 degrees C or less, average the two columns and take note that you did this in the notes column of the “TCRMP-Temperature_Benthic_SiteList&Progress” spreadsheet and in the notes column of the Temperature_UVI_deployment_log.xls. Check, process, and name the file as instructed in step 2.

If the difference is more than 0.2 degrees C, then take note of this in the “TCRMP-Temperature_Benthic_SiteList&Progress” spreadsheet and wait to process the file. Save your Excel work for reference – named, for example, as BT_TCBKPT_1509_1612_calculations.

Offset example. Proceed as with the duplicate files, but put the “b” file data set below the “a” file data set, aligning columns to match the data. Use a ‘sort’ function in the spreadsheet program to sort by the date column in ascending order. This will sequentially order the data of the two loggers. There should be a column, usually Column A, that records the number of the logging event. If the loggers came on at the same time, minus the offset, then the number of the logging events should also pair up and alternate correctly.

If there is no hobo file: take note of this and try to process the .csv file without the .hobo file. You may have to graph it in excel to see if it looks correct. Check, process, and name the file as instructed in step 2.

If there is only a hobo file: or if you are unsure that the .csv file matches the .hobo file, you can export the csv file from the hobo file.

Open the hobo file in HOBOWare. Go to file -> export table data (select *temperature only*). Check, process, and name the file as instructed in step 2.

STEP 2: CHECKING, PROCESSING, AND NAMING THE .CSV FILE

CHECK THE .CSV FILE:

Make sure start and end times in the .csv data file reflect actual site data by looking at the graph in HOBOWare (by opening the .hobo data file) to determine when the logger was deployed and recovered. When you open the graph in HOBOWare, uncheck everything except temperature.

If it does not match, or if you’re unsure, then create a new csv file from HOBOWare using the hobo file (instructions above, under “if there is only a hobo file”)

Next, check to make sure the date/time is in UTC format and not in GMT

If it is in GMT, take note of this and change the time zones to UTC (local) time. Local time is UTC -4. If the date and time are in separate columns, you will have to create a new csv file

from the hobo graph (see above: “if there is only a hobo file.”). There is a setting in HOBOWare that allows you to select the date/time format when exporting.

Also make sure that the data is in degrees *Celsius*, not *Fahrenheit*.

PROCESS THE FILE

Change the header and delete anything extraneous

Change the header information in each csv file. Copy and paste the header info from the template file “TCRMP-Temperature_Header Template”. Delete any extraneous information, such as “logger detached” from the file. The final file should have only three columns (Figure 3).

Cut off temperature data outside the deployment window

Graph the file in HOBOWare. Use the deployment log (Temperature_UVI_deployment_log.xls) to determine when the logger was emplaced and removed from the field. Before trimming the file, examine the graphed hobo file to confirm that the Date In/Time In and Date Out/Time Out columns reflect what happened in the field. Zoom in close to see where the data changes temperature abruptly (usually when it went in water or out of water), and use the icon that looks like a “+” to pinpoint the number where it changes. Double check this pinpointed value in the csv file by looking for abrupt/significant temperature changes.

Also pay attention to the temperature range - typically, there will be no temperatures above 30 or below 25, so if there are numbers way out of this range it likely means that the sensor was being lifted in/out of the water, being transported to/from the site, sitting in a air conditioned building, etc.

At the points that you choose to cut off the data (at beginning and end) delete all of the cells in excel before/after the pinpointed values. Keep the original number scale for reference (e.g. if you cut it off before sampling event #30, keep the first cell as #30), so that the final file data can always be re-associated with the original exported file.

Because this is a continuation of a deployment from the previous year it is important to avoid overlapping data between deployments. Locate and open the cleaned temperature file from the previous year’s deployment and note the date and time the data ends. Start your processed file at the next time interval after the previous year’s file stops. The previous year file should have been quality controlled and correct, but mistakes happen, so always be on the lookout for data that looks suspicious (values out of range, abrupt shifts in the temperature from time point to time point). The final result is a continuous time series of temperature data at each site.

Name the file

Update the file name to BT (stand for benthic temperature) _ six letter site code beginning with TC (found in sheet titled Site Code Metadata (see Appendix B) _ start year and month _ end year and month. This is as described for the raw data file description, but the end of deployment date is added.

Example: BT_TCFLTC_1504_1605.csv

It is important to have the file named exactly as it is in previous years (with the exception of date and logger number) based on the site names. Also make sure there are no inadvertent extra spaces or changed characters. This allows the files to be sorted in the folder and they will line correctly by date.

STEP 3: EXPORTING HOBO FILE DETAILS AND CREATING A METADATA FILE

Export details of the hobo file in HOBOWare by going to file -> export details. Name as before, but put “DETAILS” at the beginning of the file name. This should go into the folder with the processed data for the site.

Example: DETAILS_BT_TCFLTC_1504_1605.txt

Delete “coupler detached” and “end of file” sections in the text file.

Save hobo file to the final site folder, without making any changes except to make sure the end year and month are indicated. Use the same convention as before when naming the hobo file.

Example: BT_TCFLTC_1504_1605.hobo

*If you merged two serial loggers into one file, you will end up with 2 hobo files that need to be uploaded. Name the one with the lower serial number with the “_a” and higher serial number with the “_b”. E.g., BT_TCBKPT_1509_1612_a.hobo and the second BT_TCBKPT_1509_1612_b.hobo

STEP 4: CREATE THE METADATA FILE

Use the template from the file “DATASET-BT_Template” to create a metadata sheet in text editor. Save this file on the desktop to be used as a base file that will be modified for each logger deployment. Do not forget to note if you merged the files (see highlighted areas in the template for example).

Open this template into a new text file. Rename with the usual convention with “DATASET” at the beginning of the file name.

Example: DATASET_BT_TCFLTC_1504_1605.txt

The “DATASET-BT_Template” contains highlighted sections of every area that needs to be changed or updated. Use this template to follow along below regarding all of the important changes needed for each new deployment.

Change/Input:

Benthic Temperature Record: Update Full site name, location, and date of deployment

Dataset Name: Update Csv file name

Location and Depth: Update latitude/longitude and site depth; find site-specific GPS coordinates and depth in the Site Code Metadata sheet (Appendix B)

Sampling and analytical methodology: Update Serial Number(s) of HOBO loggers (found in the exported Details metadata file) and “description” and “sensor deployment info” from Site Code Metadata sheet (Appendix B). If two loggers were deployed, list both serial numbers.

Data processing: Note if files are merged: “Records from probe 1 and probe 2 were combined into a single CSV file.”

Related files and references: Copy and paste the “DETAILS” file here (from Step 3 above).

Created by: “Your Name”, date

STEP 5: UPLOAD AND UPDATE FILES

Update excel sheet named “TCRMP-Temperature_Benthic_SiteList&Progress” each time you finish a file.

Put finished files (should be 4 of them: DATA.hobo, DATA.csv, DETAILS.txt, and METADATA.txt file) in the folder titled “TCRMP Temperature Database”. There will be a folder for each site.

TROUBLESHOOTING:

“Field Notes copy” file has field notes from 2013 onward. You can use for reference if you have a question of the date of a deployment. It is fairly complete, but occasionally some notes were missed.

If you have a question that cannot be easily resolved then send the files and a detailed description of the problem to Vanessa McKague (vwright@uvi.edu) and Tyler Smith (tsmith@uvi.edu) for troubleshooting.

Remember that these files are for posterity and are used in important research and management questions. Accuracy and clarity is paramount. Rely heavily on the figures of the data and use your judgment to make sure the data makes sense (e.g., does this look like data you would expect from a site at that depth, over those seasons.). The data is getting better, but mistakes are still made in terms of site naming, etc. Your vigilance is the last line of defense in avoiding these mistakes from propagating into published data!

APPENDIX A: STANDARD OPERATING PROTOCOL FOR HOBO WATER TEMPERATURE LOGGERS

- Download and install HOBOWare:
(<http://www.onsetcomp.com/products/software/hoboware>)
- Read Hobo Logger Manual for more information regarding equipment specifications and HOBOWare User's Guide for more details. Both can be found on www.onsetcomp.com

Basic Functions in HOBOWare

- In HOBOWare, go to Preferences > General > Export Settings and update the settings as in Figure 1 below.

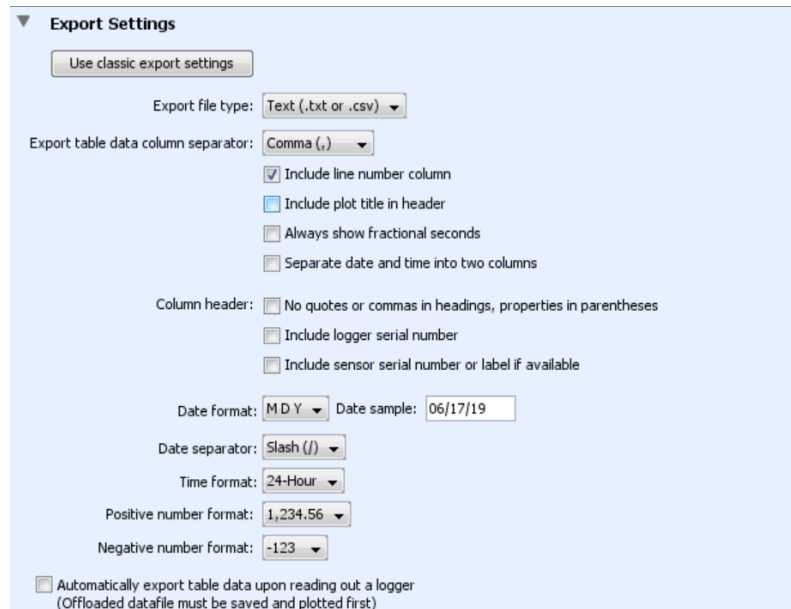


Figure 1. Export Settings in HOBOWare software that we want to standardize so that all of the exported files are similar to each other.

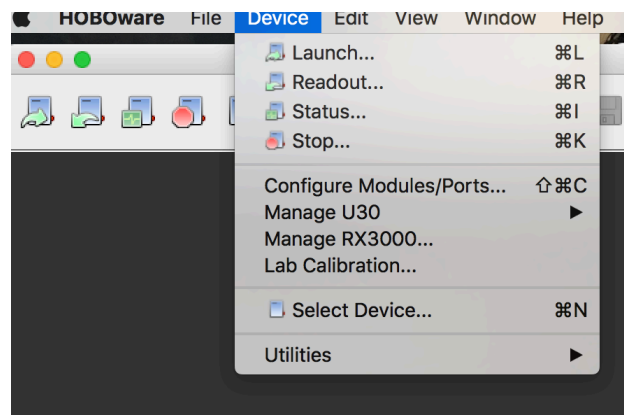


Figure 2. Device functions to launch and offload loggers.

- Figure 2 shows some of the main functions such as Launch (use when preparing a logger for the field), Readout (to offload data), Status (to check status of logger such as battery level and how times logger has been deployed).

Connecting and Deploying Loggers

- When initializing a logger all data is deleted – be sure data has been offloaded/backed up
- Open HOBOWare and plug the base station into USB port on computer
- Attach the temperature logger to the base station
- Go to the Device menu and select Launch or used the Launch icon on the toolbar
- Look at status box to verify settings and battery state
- Name the file (depending on project refer to the naming convention sheet)
- Determine logging interval (e.g., 15 minutes) and note how many days it will log until
- Select the start date and time to launch logger
- **Record all metadata (logger serial number, site location, filename, logging interval, start date/time) on an excel metadata sheet**
- Press Delayed Start button on bottom right when all information is complete
- Make sure to wrap in electrical tape and label each logger with site location

The screenshot shows the 'Launch Logger' window with the following details:

- Device:** HOB0 U22-001 Water Temp
- Name:** Calcheck_071218_10558247
- Serial Number:** 10558247
- Deployment Number:** 5
- Battery State:** GOOD (indicated by a green battery icon)
- Sensors:** 1) Temperature (checked), with a label input field containing '<Enter label here>' and a 'Filters...' button.
- Deployment:**
 - Add Interval:** A button to add a new logging interval.
 - Logging Interval Table:**

	Logging Interval	Samples	Logs until
1)	1 minute	43474	30.2 days
 - Start Logging:** On Date/Time, 07/13/18, 01:00:00 PM

Buttons at the bottom: Help, Cancel, Delayed Start.

Figure 3. Deployment settings in the Launch logger window.

Offloading and Archiving Logger Data

- Open HOBOWare and connect to logger
- Readout logger by going to Device/Readout
- Save File (a raw .hobo file is exported and can only be opened in HOBOWare)

- In Plot Setup window press Plot

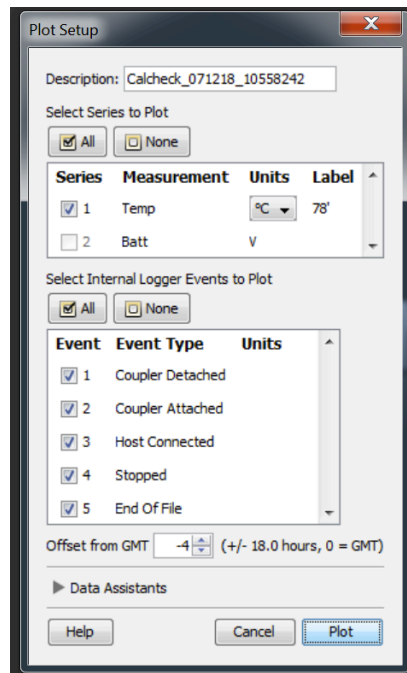


Figure 4. Plot Setup window.

- Using plotting tools and looking at the graphed data or table data determine time in water and time out of water (or more importantly make sure to have recorded this time in the field). Record this information.

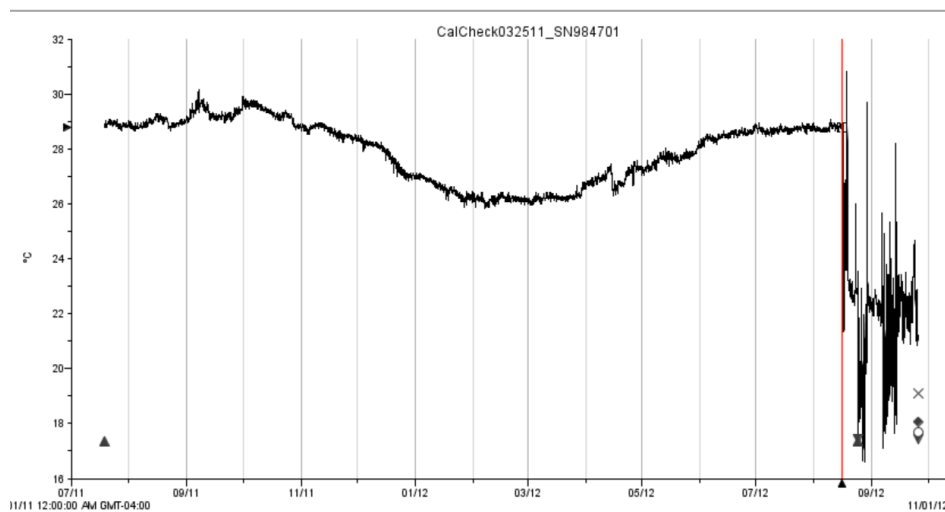


Figure 5. Plotted data with red line indicating time out of water for QC/QA of data file.

- Export the .csv file by going to File/Export Table Data (you will save a .csv file that can now be opened in Excel)
- Name .hobo and .csv file same filename so not to confuse these files, they are both the same data and need to be archived together
- Perform QA/QC on the data by clipping start/end time of deployment and cleaning up file

- Re-visit excel metadata sheet created when launching the logger and update information that logger has been removed and name of the file

Status	Location	Logger	Tag I.D.	Date On	Frequency	Duration (d)	Date In	Time In	Date Full	Date Offloaded	Offloaded Filename
Removed	Coral Bay (STJ)	10778697	na	10/24/16	15	438	10/24/16		1/5/18	2/6/18	BT_TCCORB_1610_1801.csv
Removed	St James	10776980	na	11/23/16	15	438	11/23/16		2/4/18	2/6/18	BT_TCLSTJ_1611_1801.csv
Running	Fish Bay (STJ)	10778704	na	10/24/16	15	438	10/24/16		1/5/18		

Figure 6. Example Excel sheet containing metadata of removed and running loggers.

- Some projects (i.e., TCRMP) have a more intense metadata set-up which involves an additional metadata sheet that needs to be completed and a Details file that can be saved through HOBOWare by going to File/Export Details
- Make sure all data files (raw and processed) are saved and backed up

Perform a Calibration Check on the Loggers

- All loggers (even brand new) require a calibration check before **EACH** deployment
- Fill a small cooler with ice and water and get the temperature to Zero degrees C
- Setup loggers to record at a 1 minute logging interval
- Name filename similar to this: CalCheck_YYMMDD_SerialNumber
- Stir the ice/water bath well
- Put loggers into cooler one by one making sure the sensor is submerged. The best way to get this to work is to have lots of ice in the cooler and line up each logger in the ice/water bath with the communications window facing upward, see Figure 7 below. The loggers will float but stay in position during the entire calibration check because of the ice holding them in place.



Figure 7. HOB0 U22-001 Temperature Logger showing the location of the sensor.

- Note beginning temp. of water and end temp. water by using a thermometer making sure that both temperatures are Zero degrees C (if temperature is lower that is fine, but if it is higher then that you should consider redoing the calibration check because it will be hard to determine outliers in your dataset)
- Take out the loggers after 2-4 hours
- Connect loggers to HOBOWare and offload the data (both .hobo and .csv files)
- After download look at data – if any data falls within +/- 0.2 degrees away from zero then set aside and re-do the calibration once more to confirm that the logger indeed failed the calibration check
- The loggers that passed the calibration check can be used and are ready for field deployment

Checking out and Returning Loggers

- If requesting loggers please make sure to send an email (vwright@uvi.edu), call, or stop by office of Vanessa McKague (or alternate person) at least **ONE WEEK** in advance of needing them
- If Vanessa will be preparing and launching the loggers then more time will be required and a TWO WEEK notice should be given if and when possible.
- Include a spreadsheet with all of the site information including site name and file name for each location along with the logging interval and date/time to start logging.
- Please clean and return loggers back to Vanessa in the EAL. See below on how to clean loggers.

Cleaning Loggers

- Prepare a 50/50 solution of white vinegar (household grade purchased from grocery store) and tap water in a home depot bucket or something similar
- Prepare the loggers with weights to make sure they will sink and be fully submerged in the solution
- Put loggers into the solution
- Leave them to soak overnight
- Scrub and clean the loggers and let them dry

APPENDIX B: SITE CODE METADATA

Location	Island	ReefComplex	Latitude	Longitude	Depth	6LetterCode
Coral Bay	STJ	Nearshore	18.33797	-64.70402	9	TCCORB
Fish Bay	STJ	Nearshore	18.31417	-64.76408	6	TCFSHB
Meri Shoal	STJ	Mesophotic	18.24447	-64.75862	30	TCMERI
Black Point	STT	Nearshore	18.34450	-64.98595	9	TCBKPT
Botany Bay	STT	Nearshore	18.35738	-65.03442	8	TCBOTB
Brewers Bay	STT	Nearshore	18.34403	-64.98435	6	TCBRWB
Buck Island STT	STT	Offshore	18.27883	-64.89833	14	TCBKIT
Coculus Rock	STT	Nearshore	18.31257	-64.86058	7	TCCORK
College Shoal East	STT	Mesophotic	18.18568	-65.07677	30	TCCLGE
Flat Cay	STT	Offshore	18.31822	-64.99104	12	TCFLTC
Ginsburg Fringe	STT	Mesophotic	18.18770	-64.95998	63	TCGB63
Grammanik Tiger FSA	STT	Mesophotic	18.18885	-64.95659	38	TCGMKT
Hind Bank East FSA	STT	Mesophotic	18.20217	-65.00158	39	TCHB40
Hind Bank East FSA	STT	Mesophotic	18.20217	-65.00158	30	TCHB30
Hind Bank East FSA	STT	Mesophotic	18.20217	-65.00158	20	TCHB20
Magens Bay	STT	Nearshore	18.37425	-64.93438	7	TCMAGB
Savana	STT	Offshore	18.34064	-65.08205	9	TCSAVA
Seahorse Cottage Shoal	STT	Offshore	18.29467	-64.86750	20	TCSHCS
South Capella	STT	Offshore	18.26267	-64.87237	20	TCSCAP
South Capella 35m	STT	Offshore	18.26267	-64.87237	35	TCSC35
South Water	STT	Offshore	18.28068	-64.94592	20	TCSWAT
St James	STT	Offshore	18.29459	-64.83238	15	TCLSTJ
Buck Island STX	STX	Offshore	17.78500	-64.60917	15	TCBKIX
Buck Island STX Deep	STX	Mesophotic	17.80659	-64.59935	33	TCBX33
Cane Bay	STX	Nearshore	17.77388	-64.81350	10	TCCB08
Cane Bay Deep	STX	Mesophotic	17.77661	-64.81522	38	TCCB40
Cane Bay Deep 100m	STX	Mesophotic	17.77661	-64.81522	100	TCCB99
Cane Bay Deep 67m	STX	Mesophotic	17.77661	-64.81522	67	TCCB67
Castle	STX	Offshore	17.76278	-64.59743	7	TCCSTL
Eagle Ray	STX	Offshore	17.76150	-64.69880	10	TCEAGR
Great Pond	STX	Nearshore	17.71097	-64.65221	6	TCGRPD
Jacks Bay	STX	Nearshore	17.74337	-64.57160	14	TCJCKB
Kings Corner	STX	Nearshore	17.69116	-64.90008	17	TCKNGC
Lang Bank EEMP	STX	Mesophotic	17.72145	-64.54706	27	TCLBEM
Lang Bank EEMP 100m	STX	Mesophotic	17.72145	-64.54706	100	TCLB99
Lang Bank EEMP 67m	STX	Mesophotic	17.72145	-64.54706	67	TCLB67
Lang Bank Red Hind FSA	STX	Mesophotic	17.82427	-64.44963	33	TCLBRH
Mutton Snapper FSA	STX	Offshore	17.63660	-64.86240	24	TCMT24
Mutton Snapper FSA 40m	STX	Offshore	17.63660	-64.86240	40	TCMT40

Salt River 30m	STX	Mesophotic	17.78523	-64.75917	30	TCSR30
Salt River 100m	STX	Mesophotic	17.78523	-64.75917	100	TCSR99
Salt River 41m	STX	Mesophotic	17.78523	-64.75917	41	TCSR41
Salt River 67m	STX	Mesophotic	17.78523	-64.75917	67	TCSR67
Salt River West	STX	Nearshore	17.78530	-64.75940	11	TCSR10
Sprat Hole	STX	Nearshore	17.73400	-64.89540	8	TCSPTH